

Chapter 7

Boundary value problems

7.1 General strategy

The general strategy for solving an electrostatic boundary value problem is: first, find G . Its first part is done already – we need only express it in the correct coordinates. The second part (F) requires solving the Laplace equation in the given coordinates with the boundary conditions. Second, construct ϕ from Green's theorem.

Often we get a guess at G and rely on existence/uniqueness – the method of images. If not, we bring out the special function artillery.

We encounter three types of coordinates: planar, spherical, and cylindrical.

7.2 Planar: point charge near a conducting plane

Consider a point charge q a distance b above an infinite conducting plane, on which we put $\phi = 0$ (a Dirichlet condition), as depicted in the Figure. By symmetry, we can work in the xz plane. We guess the potential by placing an image charge $-q$ a distance b below the plane

$$\phi = \frac{q}{r} - \frac{q}{r'} \quad , \quad r^2 = (b - z)^2 + x^2 \quad , \quad r'^2 = (b + z)^2 + x^2 \quad (7.1)$$

The first term solves the Poisson equation, while the second term solves the Laplace equation for $z > 0$ with the needed boundary condition. Exis-

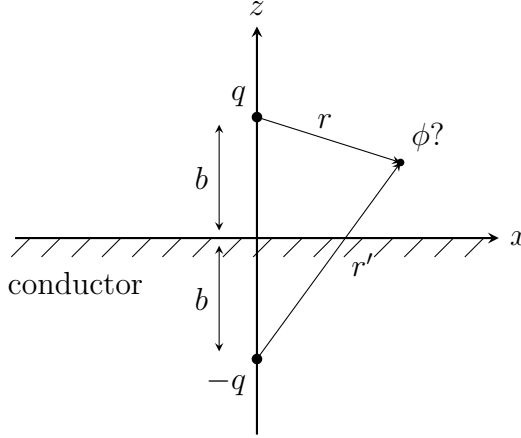


Figure 7.1: A point charge q a distance b above a conducting plane held at $\phi = 0$ (Dirichlet), with the image charge $-q$ a distance b below the plane.

tence/uniqueness then guarantees that this is the solution.

At the surface of the conductor,

$$E_z = -\frac{\partial \phi}{\partial z} \Big|_{z=0} = -\frac{2qb}{(x^2 + b^2)^{3/2}} \quad (7.2)$$

so the induced surface charge density is

$$\sigma = \frac{1}{4\pi} \left(-\frac{2qb}{(x^2 + b^2)^{3/2}} \right) \quad (7.3)$$

The total induced charge is then

$$\int \sigma da = -\frac{qb}{2\pi} \iint \frac{r dr d\theta}{(r^2 + b^2)^{3/2}} = -q = \text{image charge} \quad (7.4)$$

7.3 2D semi-infinite rectangle

Consider a two-dimensional semi-infinite rectangle with $\rho = 0$ inside: $\nabla^2 \phi = 0$ in the interior, with $\phi = 0$ on the two sides at $x = 0$ and $x = b$, and $\phi = \phi_0$ on the bottom at $z = 0$, as depicted in the Figure. Since $\rho = 0$, we only need F ; and we might as well solve for ϕ directly – $\nabla^2 \phi = 0$ with the given

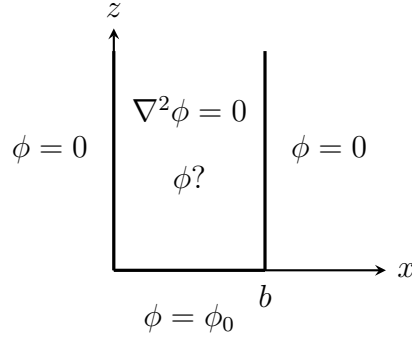


Figure 7.2: A two-dimensional semi-infinite rectangle with no charge inside: $\phi = 0$ on the two sides, $\phi = \phi_0$ on the bottom.

boundary conditions – instead of getting F , then G , then applying Green's theorem.

In Cartesian coordinates, we separate variables

$$\phi(x, y, z) = X(x) Y(y) Z(z) \quad (7.5)$$

so that

$$\nabla^2 \phi = 0 \Rightarrow \frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = 0 \quad (7.6)$$

Each term must separately be a constant, which we call $-\alpha^2$, $-\beta^2$, and γ^2 respectively – simple harmonic oscillator equations in x and y , exponentials in z – with $\gamma^2 = \alpha^2 + \beta^2$. Hence

$$\phi = e^{\pm i \alpha x} e^{\pm i \beta y} e^{\pm \sqrt{\alpha^2 + \beta^2} z} \quad (7.7)$$

We pick z as the exponential direction because of the boundary conditions. Translational symmetry in y gives $\beta = 0$, and

$$\phi(0, y, z) = \phi(b, y, z) = 0 \Rightarrow X = \sin\left(\frac{n\pi x}{b}\right) \quad (7.8)$$

Hence

$$\phi = \sum_{n=1}^{\infty} A_n e^{-n\pi z/b} \sin\left(\frac{n\pi x}{b}\right) \quad (7.9)$$

with the coefficients fixed by the boundary condition $\phi(x, 0, 0) = \phi_0$ at the bottom wall

$$\begin{aligned} A_n &= \frac{2}{b} \int_0^b \phi(x, 0, 0) \sin\left(\frac{n\pi x}{b}\right) dx \\ &= \frac{4\phi_0}{\pi n} \begin{cases} 1 & n \text{ odd} \\ 0 & n \text{ even} \end{cases} \end{aligned} \quad (7.10)$$

so that

$$\phi = \frac{4\phi_0}{\pi} \sum_{n \text{ odd}} e^{-\pi n z/b} \sin\left(\frac{n\pi x}{b}\right) \quad (7.11)$$

Note that the series can be resummed: take

$$\xi = e^{i\frac{\pi}{b}(x+iz)} \quad (7.12)$$

so that

$$\begin{aligned} \phi &= \frac{4\phi_0}{\pi} \operatorname{Im} \sum_{n \text{ odd}} \frac{\xi^n}{n} = \frac{2\phi_0}{\pi} \operatorname{Im} \ln \frac{1+\xi}{1-\xi} \\ &= \frac{2\phi_0}{\pi} \tan^{-1} \frac{\sin \pi x/b}{\sinh \pi z/b} \end{aligned} \quad (7.13)$$

Generally, we want

$$G = \frac{1}{|\vec{x} - \vec{x}'|} + F \quad (7.14)$$

where the first term is written in Cartesian coordinates – though it can also be written in Fourier modes – and F solves the Laplace equation in Cartesian coordinates, to be decomposed in Fourier modes.

7.4 Review of orthogonal special functions

We expand a function on an interval in a set of orthonormal functions

$$f(x) = \sum_{n=1}^{\infty} a_n U_n(x) \quad , \quad a < x < b \quad (7.15)$$

with

$$\int_a^b U_n(x) U_m^*(x) dx = \delta_{nm} \Rightarrow a_m = \int_a^b f(x) U_m^*(x) dx \quad (7.16)$$

The $U_n(x)$ often arise as solutions to a PDE – a link to group theory. For example,

$$U_n(x) = \frac{1}{\sqrt{\pi}} \sin(nx) \quad , \quad n = 1, 2, \dots \quad , \quad -\pi \leq x \leq \pi \quad (7.17)$$

for odd functions. More generally, for periodic functions,

$$U_n = \sqrt{\frac{2}{d}} \sin \frac{2\pi nx}{d} \quad , \quad V_n = \sqrt{\frac{2}{d}} \cos \frac{2\pi nx}{d} \quad , \quad -\frac{d}{2} \leq x \leq \frac{d}{2} \quad (7.18)$$

But these can also be used with non-periodic functions on an interval using the “doubling trick”: extend

$$f(x+d) = f(d-x) \quad (7.19)$$

so that $f(x)$ is defined over $0 \leq x \leq 2d$.

In the continuum, we have Fourier transforms, with $u(k, x) = e^{ikx}/\sqrt{2\pi}$:

$$f(x) = \int_{-\infty}^{\infty} A(k) \frac{e^{ikx}}{\sqrt{2\pi}} dk \quad , \quad A(k) = \int_{-\infty}^{\infty} f(x) \frac{e^{-ikx}}{\sqrt{2\pi}} dx \quad (7.20)$$

with orthonormality

$$\delta(k - k') = \int_{-\infty}^{\infty} \frac{e^{ikx}}{\sqrt{2\pi}} \left(\frac{e^{ik'x}}{\sqrt{2\pi}} \right)^* dx \quad (7.21)$$